

Suggested FYP topics for 2026-27

by

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Project code :	HC1
Project title :	Cross-Layer Wireless Intelligence for Connected Robotics
No. of students :	2
Description of project :	Today's robots often treat their wireless interfaces, such as WiFi, as "black boxes." This approach leads to highly variable connectivity during motion, which significantly degrades performance when robots need to offload computing tasks to edge servers or be teleoperated by remote operators. To address this problem, this project aims to empower future robots with cross-layer wireless intelligence. By enabling robots to perceive lower-layer insights embedded within their WiFi interfaces, such as transmission (Tx) information and channel state information (CSI), the system can agilely and jointly optimize both its physical motion and data transmission to boost end-to-end performance. Furthermore, robotic network security is a critical concern; a remotely hijacked robot poses a severe risk of causing physical harm. Therefore, students participating in this project are also encouraged to explore how to leverage these lower-layer insights to detect threats and enhance the network security of robotic systems. Expectations: Good Linux-based programming skills, alongside a good foundation in computer networks and cybersecurity.
Industrial collaboration :	No.

Project code :	HC2
Project title :	Multimodal-Sensory Intelligence for Robotic Manipulation
No. of students :	2
Description of project :	Today's robots primarily rely on cameras to perceive their environment for object manipulation tasks. However, standard RGB-D cameras struggle when encountering transparent objects, liquids, or visual occlusions. Furthermore, perceiving material properties, which is essential for robust manipulation, often falls beyond the sensing capabilities of traditional cameras. To address these challenges, this project aims to equip robots with multimodal sensory intelligence. Specifically, we will develop a comprehensive sensor platform integrating an RGB-D camera, a thermal camera, and an mmWave radar. Leveraging this hardware platform, the project will develop multimodal LLM-based agentic solutions to adaptively fuse these diverse sensing modalities, achieving robust perception across complex situations. Expectations: Good foundation in signal processing and practical experience with multimodal LLM.

Industrial collaboration :	No.
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